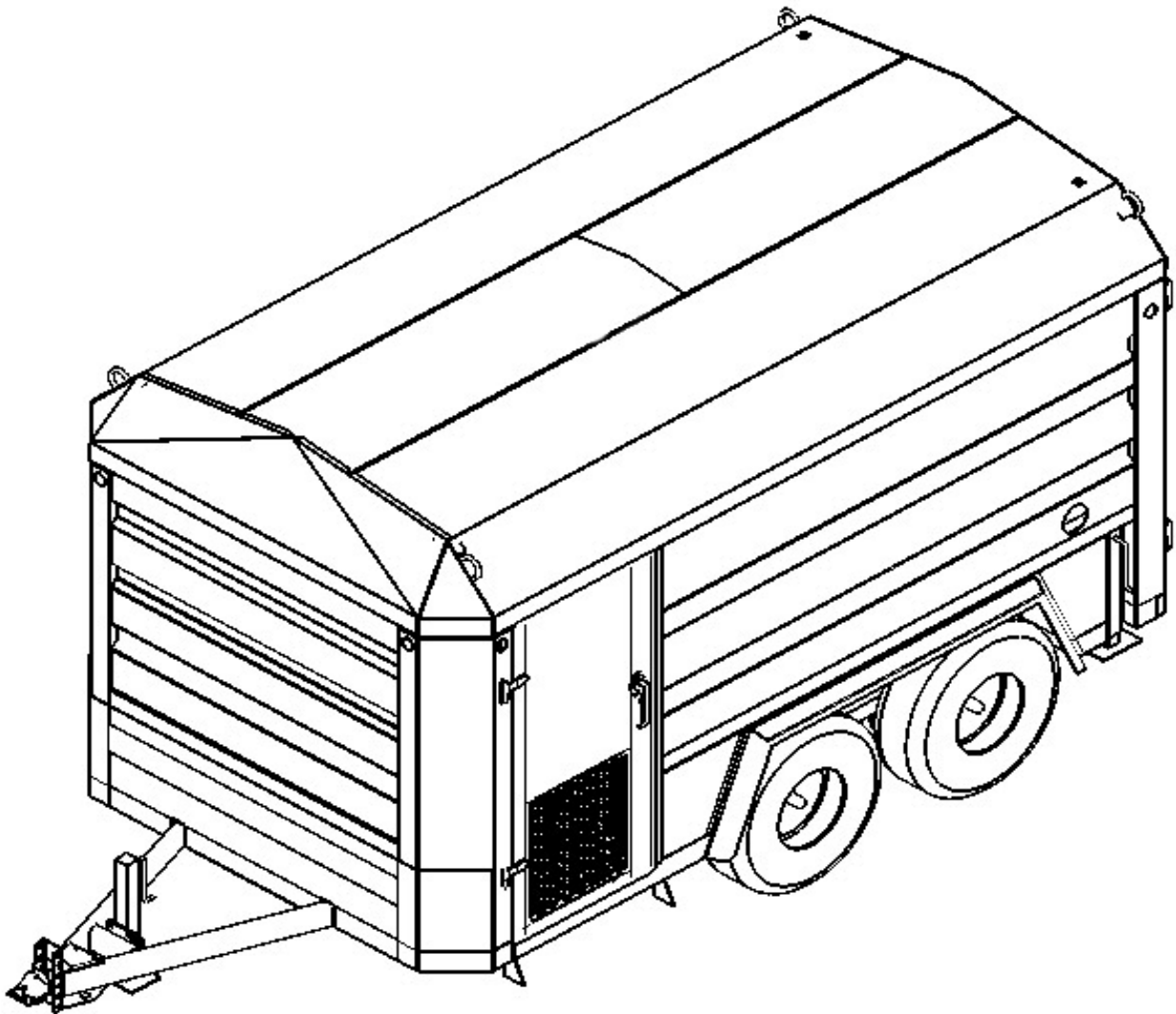


REPAIR PARTS MANUAL

**SERIES
BULLETIN No.
SUPERSEDES**

HEAT KING ~ HK500

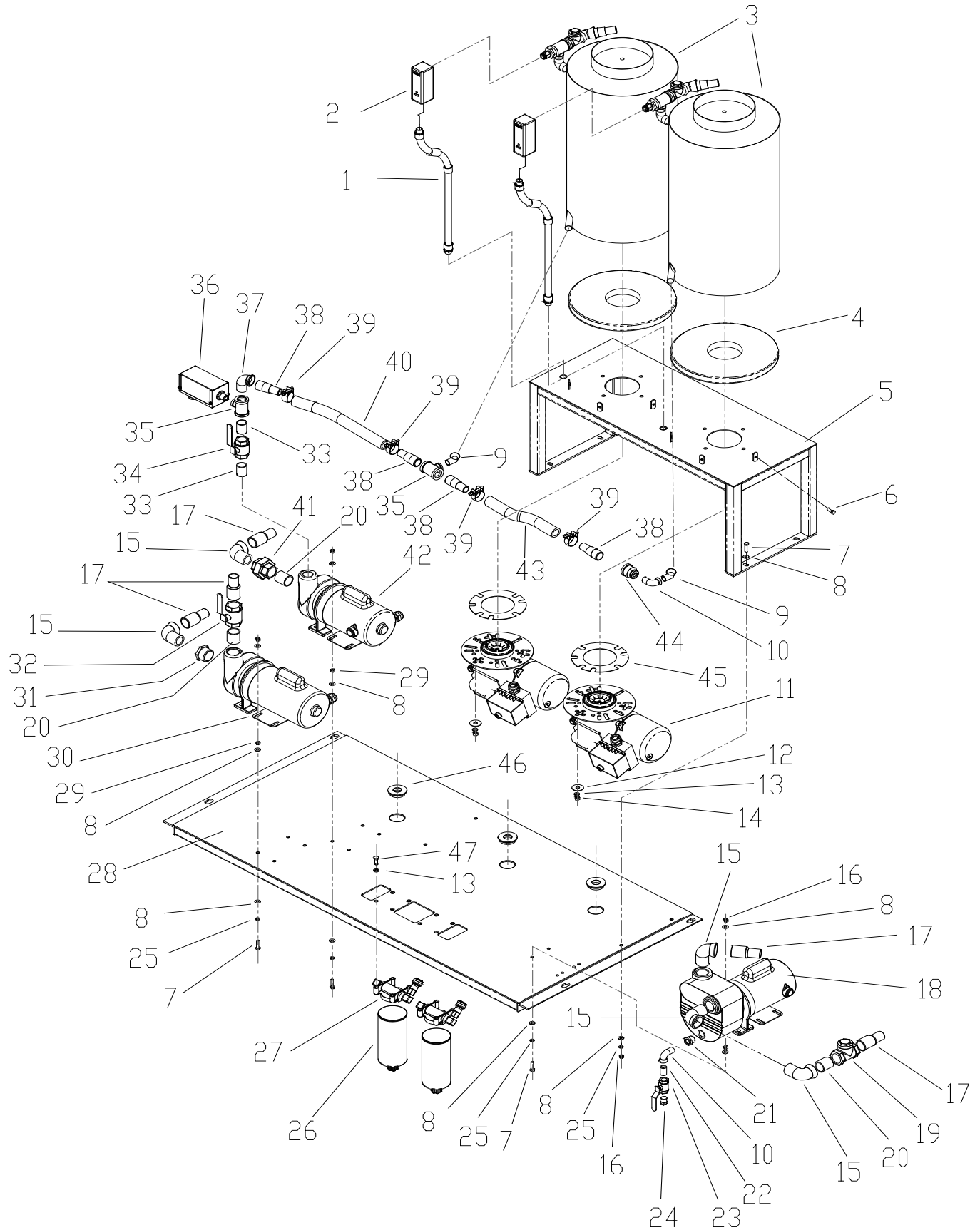
Mobile Glycol Heating System



CROWN
CONSTRUCTION EQUIPMENT

Printed
in
CANADA

Pump Plate Assembly

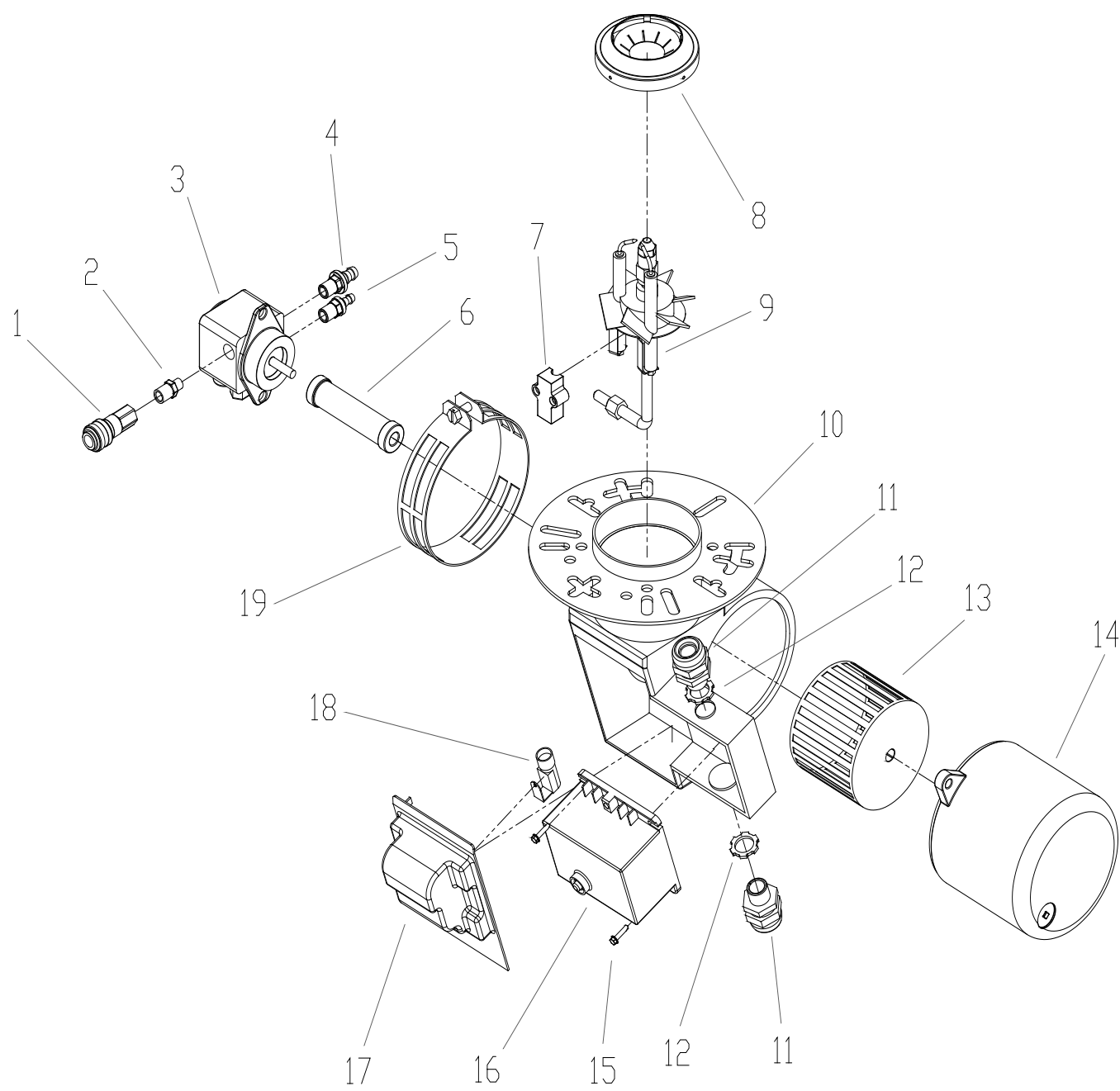


Pump Plate Assembly

Ref	Part	Description	Qty
1	446605	PVC Conduit Assembly	2
2	446542	Aquastst, High Limit	2
3	446608	Coil Assembly	2
4	441135	Insulation, Burner	2
5	440188	Coil Base Weldment	1
6	225980	Bolt, HexHd, 5/16 x 1	6
7	225990	Bolt, HexHd, 5/16 x 1-1/4	16
8	174990	Washer, 5/16	44
9	444742	Elbow, Street, 1/2 x 45°, Galv.	2
10	444610	Elbow, Street, 1/2, Galv.	2
11	446606	Burner Assembly	2
12	174987	Washer,	8
13	176230	Washer, 3/8, Spring	16
14	227830	Nut, Hex, 3/8	8
15	444250	Elbow, Street, 1-1/4 x 90°, Galv.	5
16	227821	Nut, Hex, 5/16, Nylon Lock	16
17	444330	Barb, Hose, 1-1/4, Galv.	5
18	446603	Fill Pump Assembly	1
19	444630	Valve, Swing Check, 1-1/4.	1
20	444480	Nipple, Close, 1-1/4, Galv.	3
21	444085	Bushing Reducer, 3/4 x 1/2 NPT	1
22	444440	Nipple, Close, 1/2, Galv.	1
23	444120	Valve, Ball, 1/2	1
24	444083	Plug, 1/2, Galv.	1

Ref	Part	Description	Qty
25	176220	Washer, 5/16, Spring	16
26	445157	Fuel Filter	2
27	446607	Fuel Filter Head Assembly	2
28	440131	Pump Plate Weldment	1
29	227820	Nut, Hex, 5/16	12
30	446601	Field Pump Assembly	1
31	444749	Bushing, 1-1/2 x 1-1/4, Galv.	1
32	444110	Valve, Ball, 1-1/4	1
33	444740	Nipple, Close, 1, Galv.	2
34	444819	Valve, Ball, 1, Brass	1
35	444821	Tee, 1 x 1 x 1/2, Galv.	2
36	446604	Pressure Switch Assembly	1
37	444820	Elbow, Street, 1 x 90°, Galv.	1
38	444822	Barb, Hose, 1, Galv.	4
39	444810	Clamp, 1, T-Bolt	4
40	446611	Hose, Circ Pump to Coil	1
41	444763	Union, 1-1/4, Galv.	1
42	446602	Circulation Pump Assembly	1
43	446612	Hose, Coil to Coil	1
44	444816	Bell Reducer, 1 x 1/2, Galv.	1
45	441136	Gasket Seal	2
46	440203	Grommet, Coil Plate	3
47	226210	Bolt, HexHd, 3/8 x 1	8

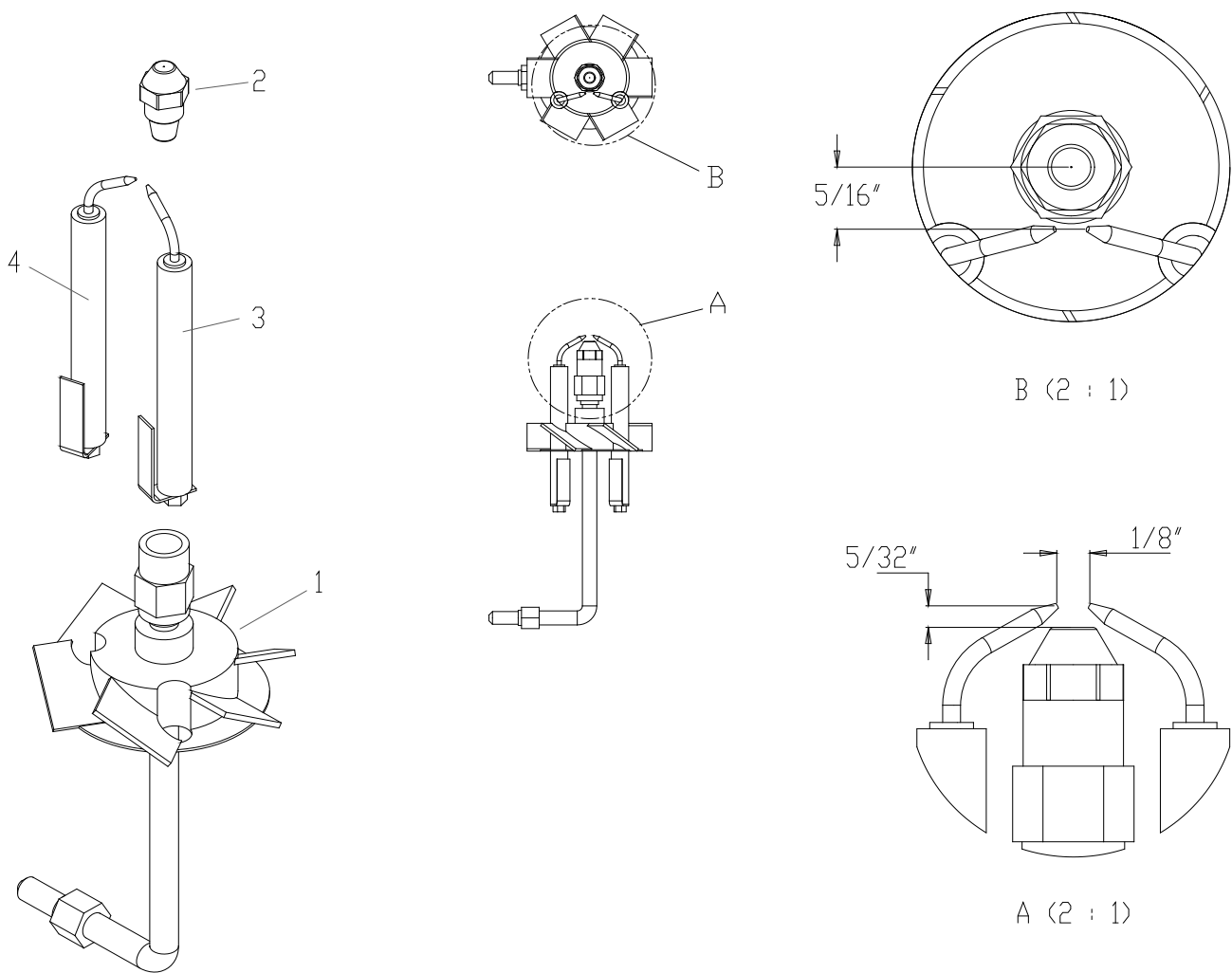
Burner Assembly



Burner Assembly

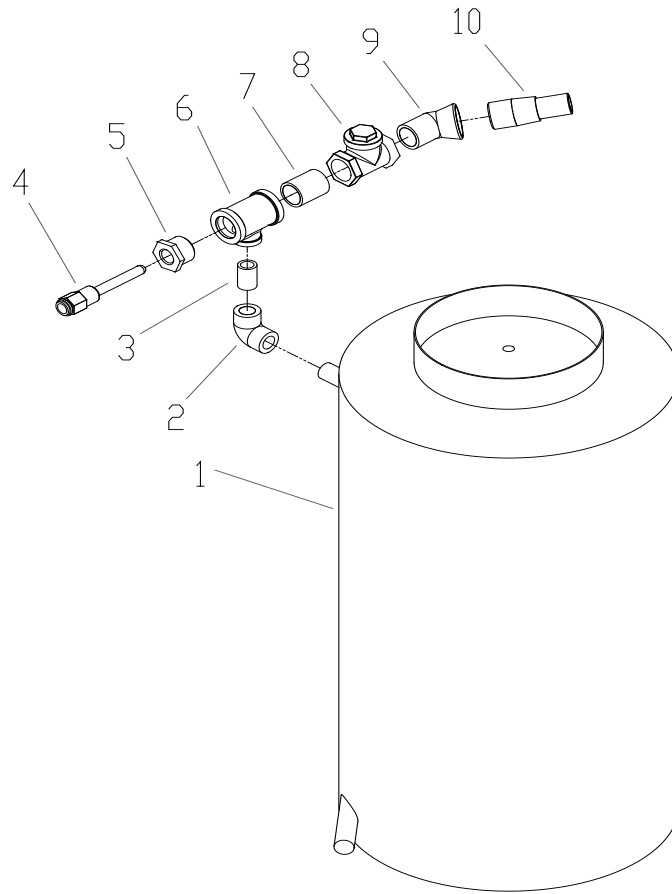
Ref	Part	Description	Qty
1	444032	Quick Disconnect, 1/4, Female	1
2	444814	Adapter, 1/8 x 1/4 Male	1
3	449001	Fuel Pump, 3450 rpm, 3 gph	1
4	444021	Nipple, Hose, 1/4 x 3/8, Brass	1
5	444042	Nipple, Hose, 1/4 x 1/4, Brass	1
6	449032	Flexible Coupling	1
7	446544	Preheater	1
8	449037	Reten Head	1
9	449013	Drawer Assembly	1
10	449048	Burner Chassis	1
11	446260	Connector, 1/2, Stress Relief	2
12	112510	Nut, 1/2, Lock, Conduit	2
13	449028	Fan, Squirrel Cage	1
14	449024	Fan Motor	1
15	162830	Screw, 8-18 x 3/4	2
16	446490	Sensor, Primary Control Relay	1
17	449014	Transformer, Igniter	1
18	446491	Sensor, Cad Cell Lead	1
19	446768	Air Band	1

Burner Drawer Assembly



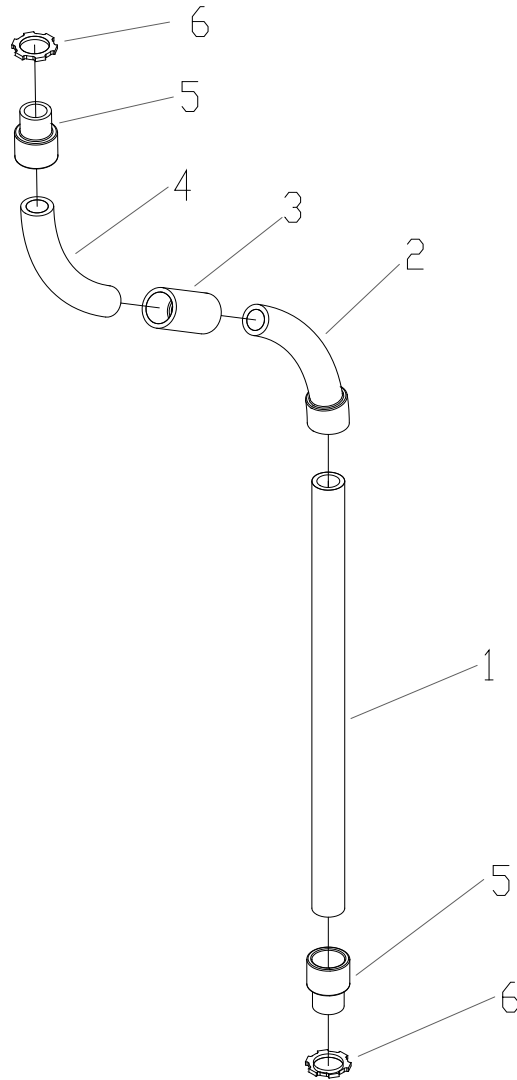
Ref	Part	Description	Qty
1	446615	Alkota Burner Drawer	1
2	449212	Nozzle, 1.65 gph 80D A Type	1
3	449003	Burner, Electr R V-100772-001	1
4	449030	Burner, Electr L V-100773-001	1

Coil Assembly



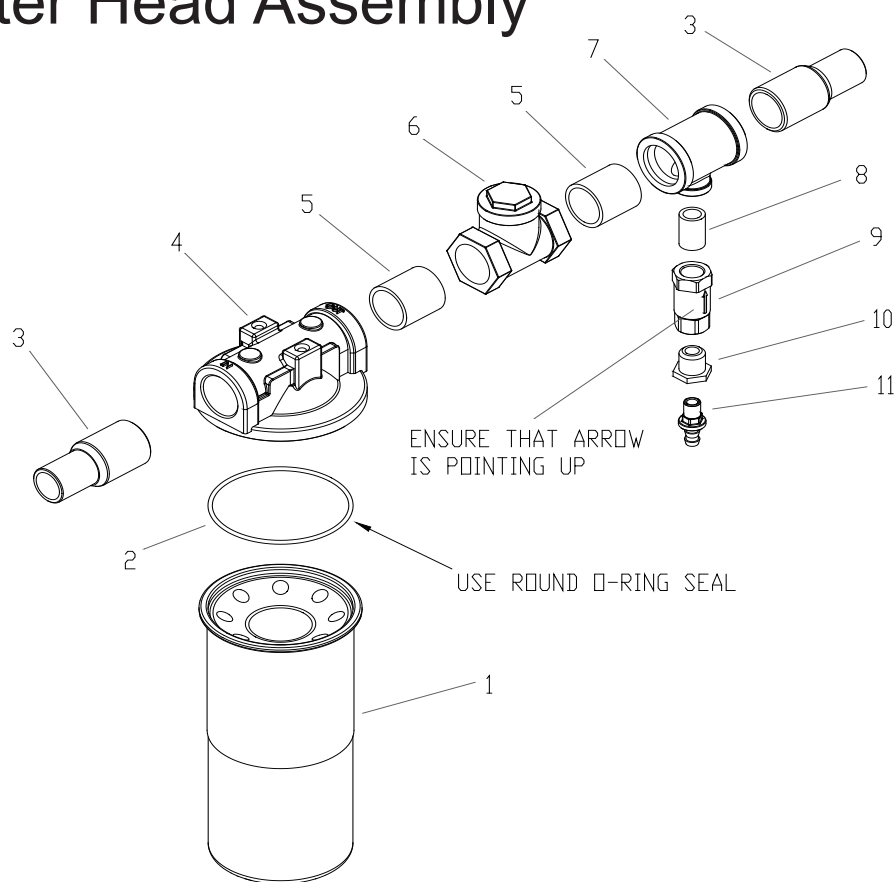
Ref	Part	Description	Qty
1	441137	Coil, 2122-00207	1
2	446610	Elbow 1/2 x 90°, Alkota	1
3	444440	Nipple, Close, 1/2, Galv.	1
4	446543	High Limit, Aquastat Well	1
5	444817	Bushing, Reducer, 1 x 1/2, Galv.	1
6	444821	Tee, 1 x 1 x 1/2, Galv.	1
7	444740	Nipple, Close, 1, Galv.	1
8	444811	Check Valve, 1, Swing, Brass.	1
9	444827	Elbow 1 x 45°, Street, Galv.	1
10	444822	Barb, Hose, 1, Galv.	1

PVC Conduit Assembly



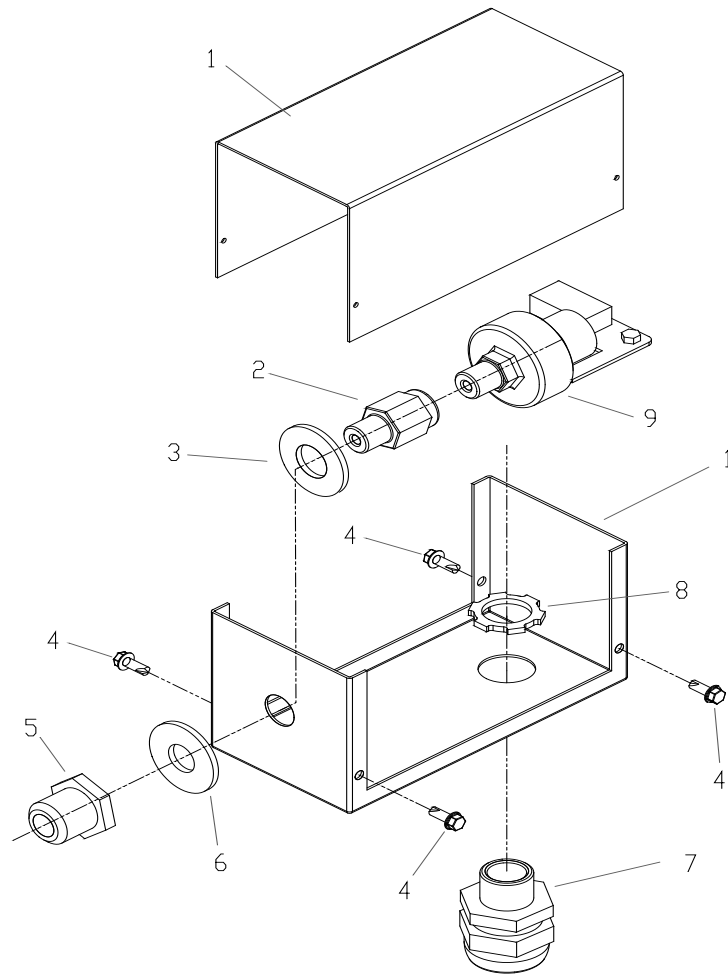
Ref	Part	Description	Qty
1	446773	Conduit, High Limit	1
2	446771	Elbow, Lower High Limit	1
3	446041	Coupler, 1/2, PVC	1
4	446772	Elbow, Upper High Limit	1
5	446537	Adapter, Terminal, 1/2, PVC	2
6	112510	Nut, Lock, 1/2, Conduit	2

Glycol Filter Head Assembly



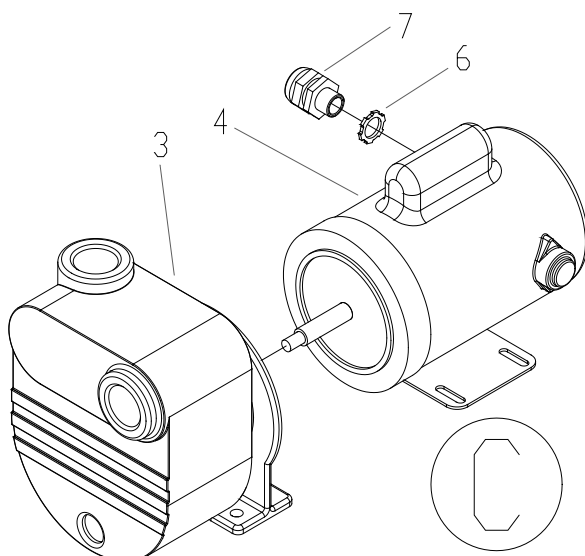
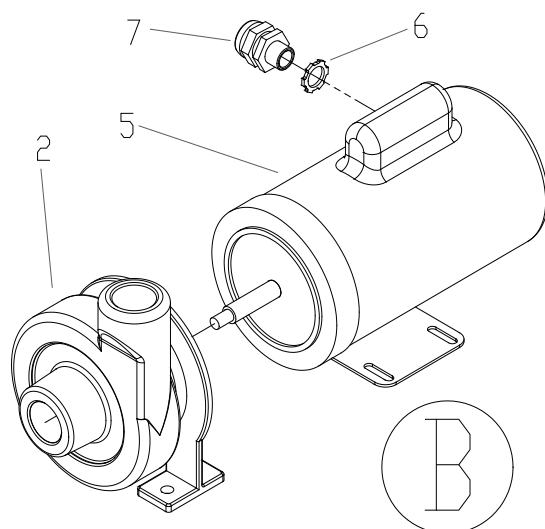
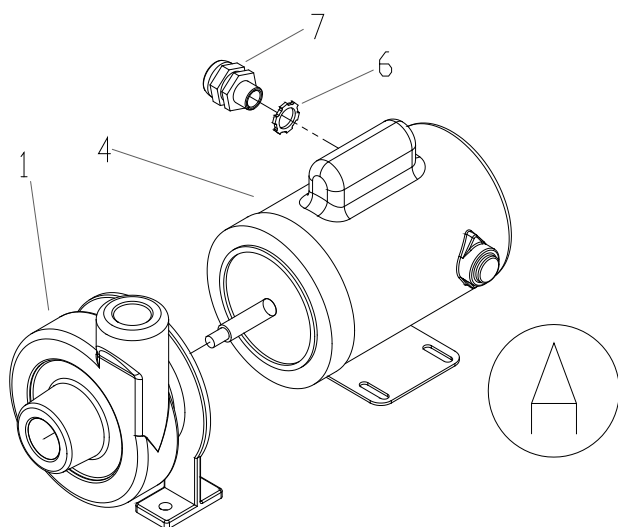
Ref	Part	Description	Qty
1	445159	Filter Cartridge	1
2	446746	O-Ring Seal	1
3	444330	Barb, Hose, 1-1/4, Galv.	2
4	445158	Filter Assembly	1
5	444480	Nipple, Close, 1-1/4, Galv.	2
6	444630	Valve, Swing Check, 1-1/4	1
7	444540	Tee, Reducer, 1-1/4 x 1-1/4 x 2	1
8	444440	Nipple, Close, 1/2, Galv.	1
9	444731	Valve, Inline Check, 1/2, Brass	1
10	444832	Bushing, 1/2 x 1/4, Galv.	1
11	444021	Nipple, Hose, 1/4 NPT x 3/8, Brass	1

Pressure Switch Assembly



Ref	Part	Description	Qty
1	446553	Case	1
2	446523	Pressure Switch, Snubber	1
3	445514	Washer, Pressure Switch, Large	1
4	162832	Screw, 8-18 x 1/2	4
5	444832	Bushing, 1/2 x 1/4, Glav.	1
6	446613	Washer, Pressure Switch, Small	1
7	446260	Connector, 1/2, Stress Relief	1
8	112510	Nut, Lock 1/2, Conduit	1
9	447011	Pressure Switch	1

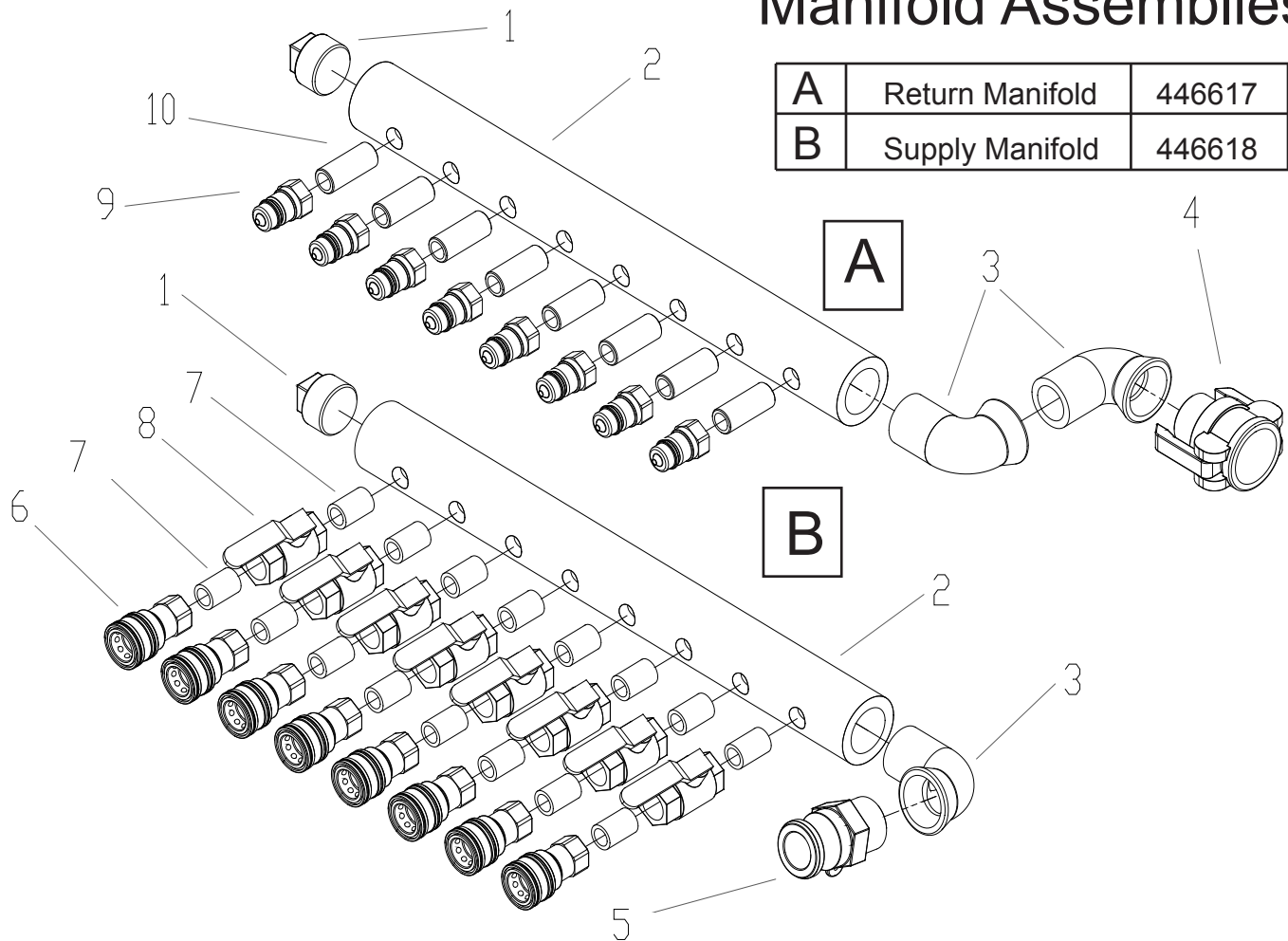
Pump~Motor Information



A	Field Pump Assy	446601
B	Circ. Pump Assy	446602
C	Fill Pump Assy	446603

Ref	Part	Description	Qty
1	441180	Circulating Pump, 1/3 Hp	1
2	441182	Field Pump, 3/4 Hp	1
3	441183	Fill Pump, 1/3 Hp	1
4	219110	Motor, 1/3 Hp, 1-115/230V	2
5	219106	Motor, 3/4 Hp, 1-115/230V	1
6	112510	Nut, Lock, 1/2, Conduit	3
7	446260	Connector, 1/2, Stress Relief	3

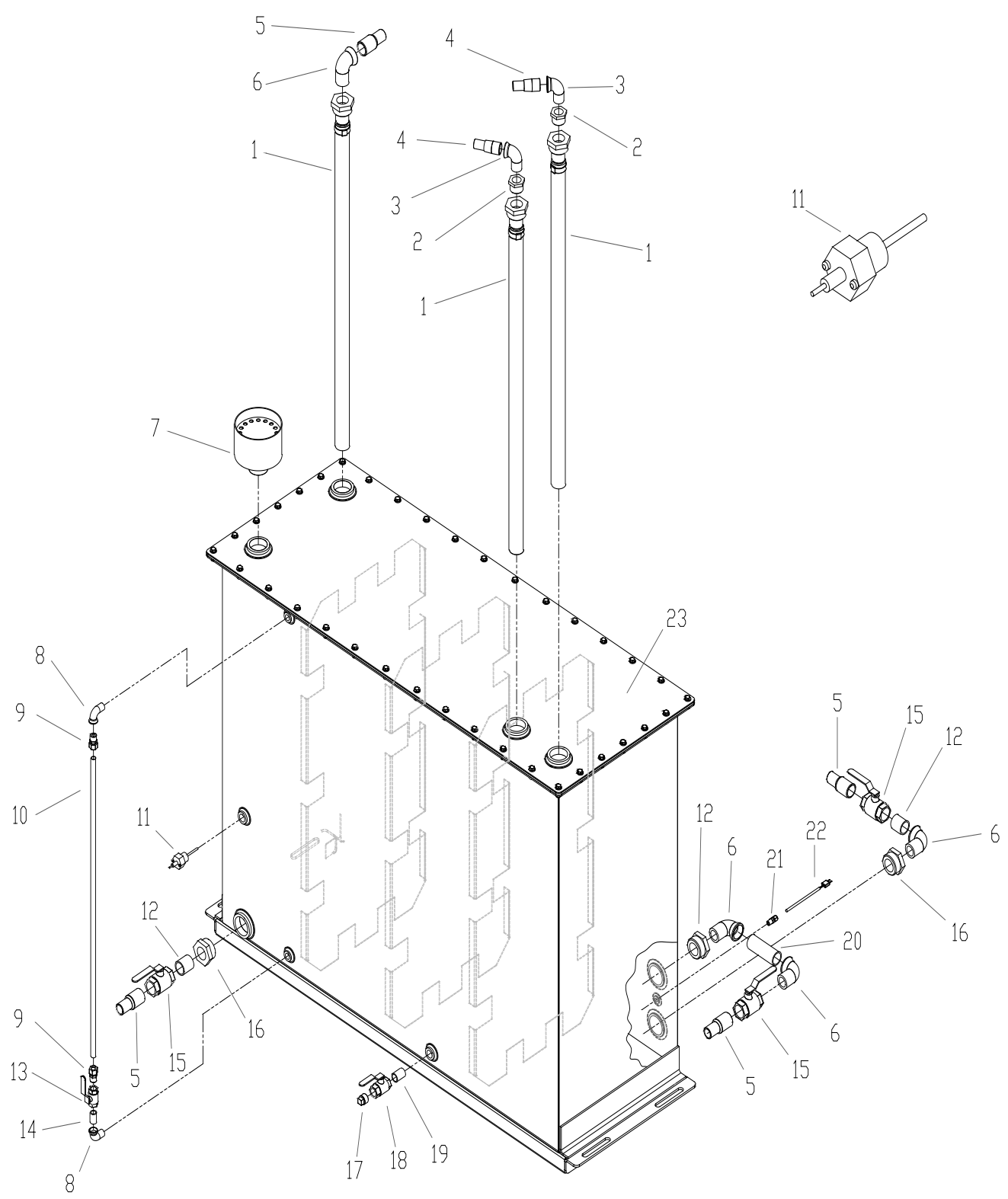
Manifold Assemblies



A	Return Manifold	446617
B	Supply Manifold	446618

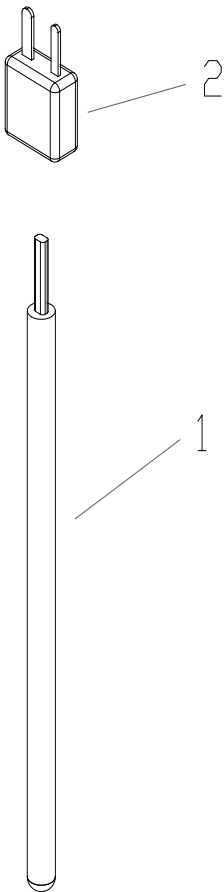
Ref	Part	Description	Qty
1	444089	Plug, 1-1/4 NPT, Galv.	2
2	444808	Manifold, 8 Port	2
3	444250	Elbow, 1-1/4 x 90°, Street, Galv.	3
4	444751	Camlock, 1-1/4 NPT Male x 1-1/4 Fem.	1
5	444754	Camlock, 1-1/4 NPT Male x 1-1/4 Male.	1
6	444290	Quick Connect, 1/2, Female	8
7	444440	Nipple, Close, 1/2, Galv.	16
8	444081	Valve, Ball, 3/4 NPT, Brass	8
9	444400	Quick Connect, 1/2, Male	8
10	444805	Nipple, 1/2 x 2, Galv.	8

Glycol Tank Assembly



Ref	Part	Description	Qty
1	446701	Glycol Return Tube Assembly	3
2	444818	Reducer, 1-1/4 x 1, Galv.	2
3	444820	Elbow, Street, 1 ~ 90°, Galv.	2
4	444822	Hose Barb, 1, Galv.	2
5	444330	Hose Barb, 1-1/4, Galv.	4
6	444250	Elbow, Street, 1-1/4 ~ 90° Galv.	4
7	444704	Breather, 500MBTU, Assembly	1
8	444610	Elbow, Street, 1/2, Galv.	2
9	444200	Ftg, Compression, 1/2 NPT x 5/8	2
10	442160	Poly Tubing, 1/2 i/d x 5/8 o/d	1
11	441140	Low Level Cut Off Probe and Switch	1
12	444480	Nipple, Close, 1-1/4, Galv.	2
13	444120	Ball Valve, 1/2	1
14	444440	Nipple, Close, 1/2, Galv.	1
15	444110	Ball Valve, 1-1/4	3
16	444140	Bushing, 2 x 1-1/4, Galv.	3
17	444082	Plug, 3/4 NPT, Galv.	1
18	444081	Ball Valve, 3/4, Brass	1
19	444460	Nipple, Close, 3/4, Galv.	1
20	444804	Nipple, 1-1/4 x 4-1/2 Galv.	1
21	444758	Ftg, Compression, 1/2 NPT x 5/8	1
22	446704	Thermocouple Assembly	1
23	446705	Tank Assembly	1

Thermocouple Assembly

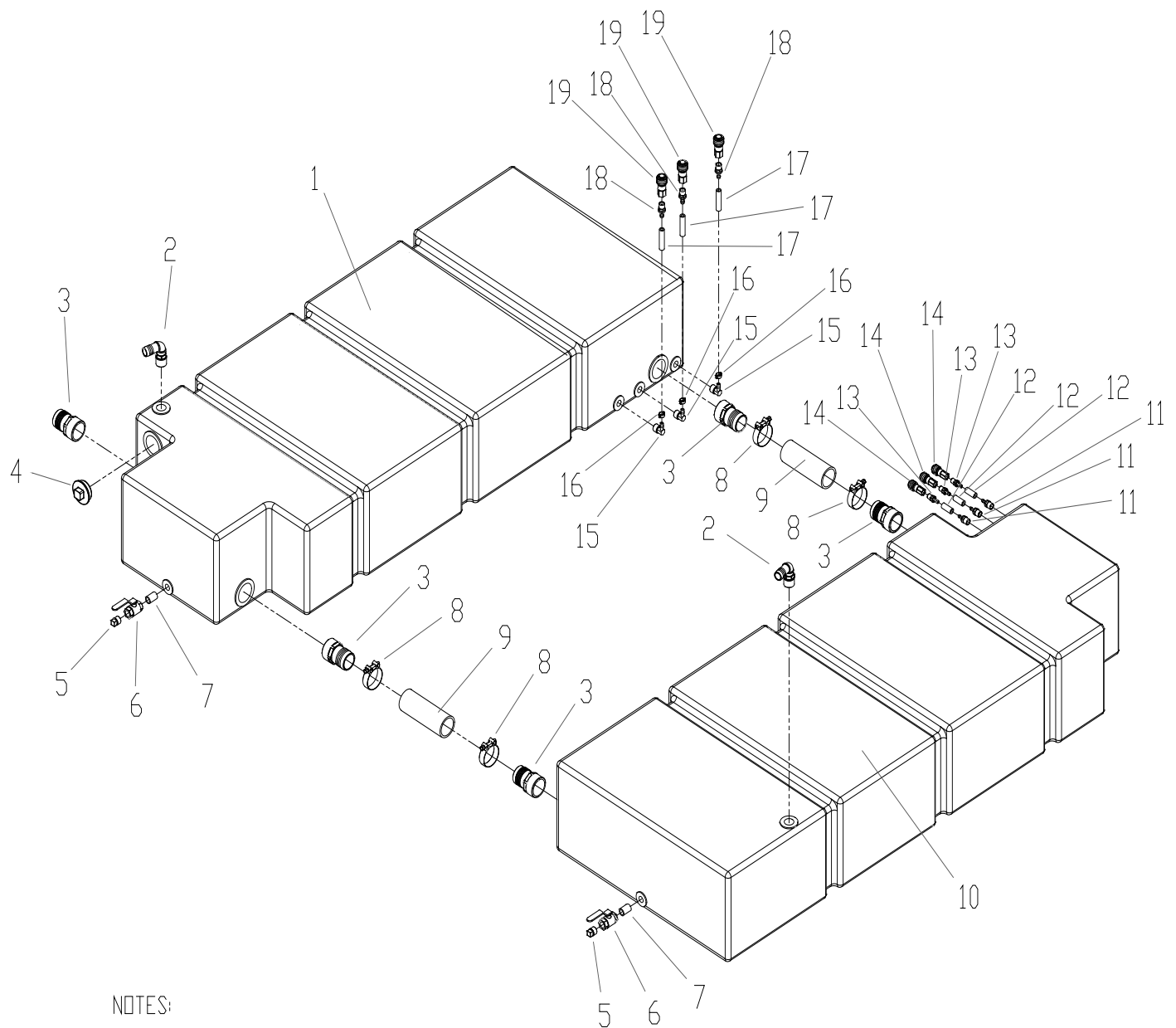


NOTE:

CONNECT BLUE WIRE TO "+" ON CONNECTOR
CONNECT RED WIRE TO "-" ON CONNECTOR

Ref	Part	Description	Qty
1	446460	Thermocouple, HTTC36-T-14G-6	1
2	446469	Connector, Blue, T/C Male	1

Fuel Tank Assembly



NOTES:

FOR FITTINGS THAT CONTACT POLY USE:

- LOCTITE 7649 PRIMER
- LOCTITE 567 PIPE DOPE

FOR FITTINGS THAT CONTACT BRASS TO BRASS USE:

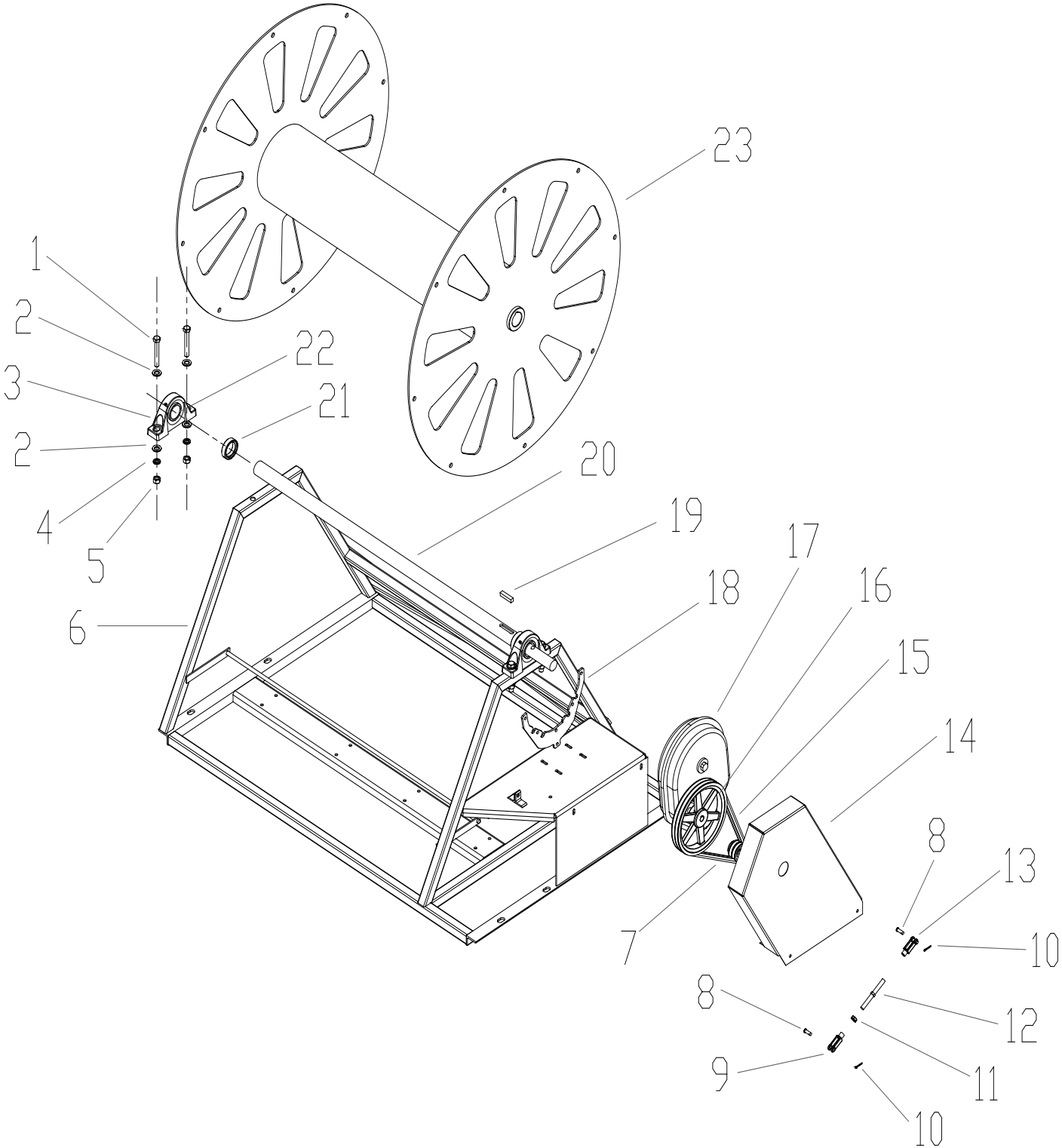
- LOCTITE 567 PIPE DOPE

FOR FUEL CROSSOVER 2" PVC FITTINGS THAT LEAK
IN THE FIELD REPLACE 444828 WITH 446726

Fuel Tank Assembly

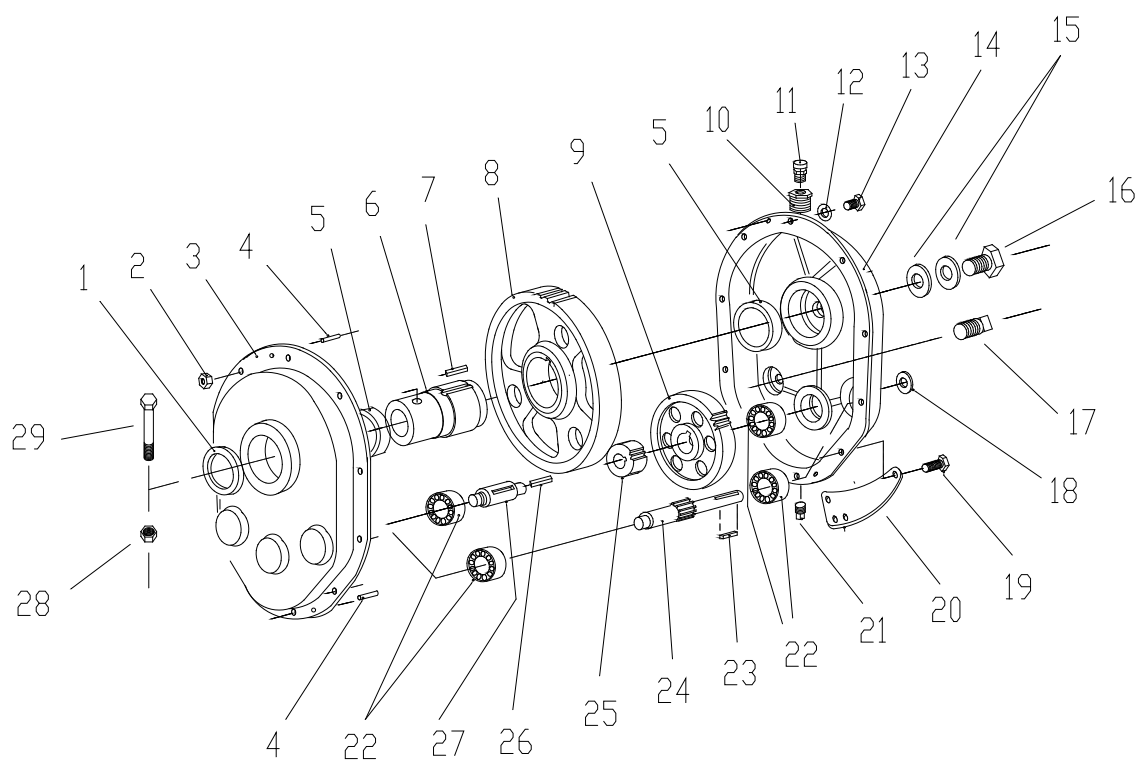
Ref	Part	Description	Qty
1	441225	Tank, Left	1
2	444829	Elbow, 90°, 1 NPT x 1 PVC	2
3	444828	Nipple, Hose, 2 NPT x 2 PVC	5
4	444831	Plug, 2 PVC	1
5	444083	Plug, 1/2 NPT, Galv.	2
6	444120	Ball Valve, 1/2.	2
7	444440	Nipple, Close, 1/2, Galv.	2
8	444834	Clamp, 2	4
9	446737	Hose, 2, Fuel Cross Over	2
10	441226	Tank, Right	1
11	444812	Nipple, 3/8 NPT x 1/4 LOL, Brass	3
12	446743	Hose, 1/4 LOL x 28 long	3
13	444042	Nipple, 1/4 NPT x 1/4 Hose, Brass	3
14	444032	1/4 Quick Disconnect - Female	3
15	444815	Nipple, 90°, 1/2" NPT x 3/8 Hose	3
16	444825	Clamp, 3/8, - Otiker FVHC9-9-1/2	3
17	446742	Hose, 3/8 LOL x 28 long	3
18	444022	Nipple, 3/8 NPT x 3/8 LOL, Brass	3
19	444024	3/8 Quick Disconnect - Female	3

Reel Assembly



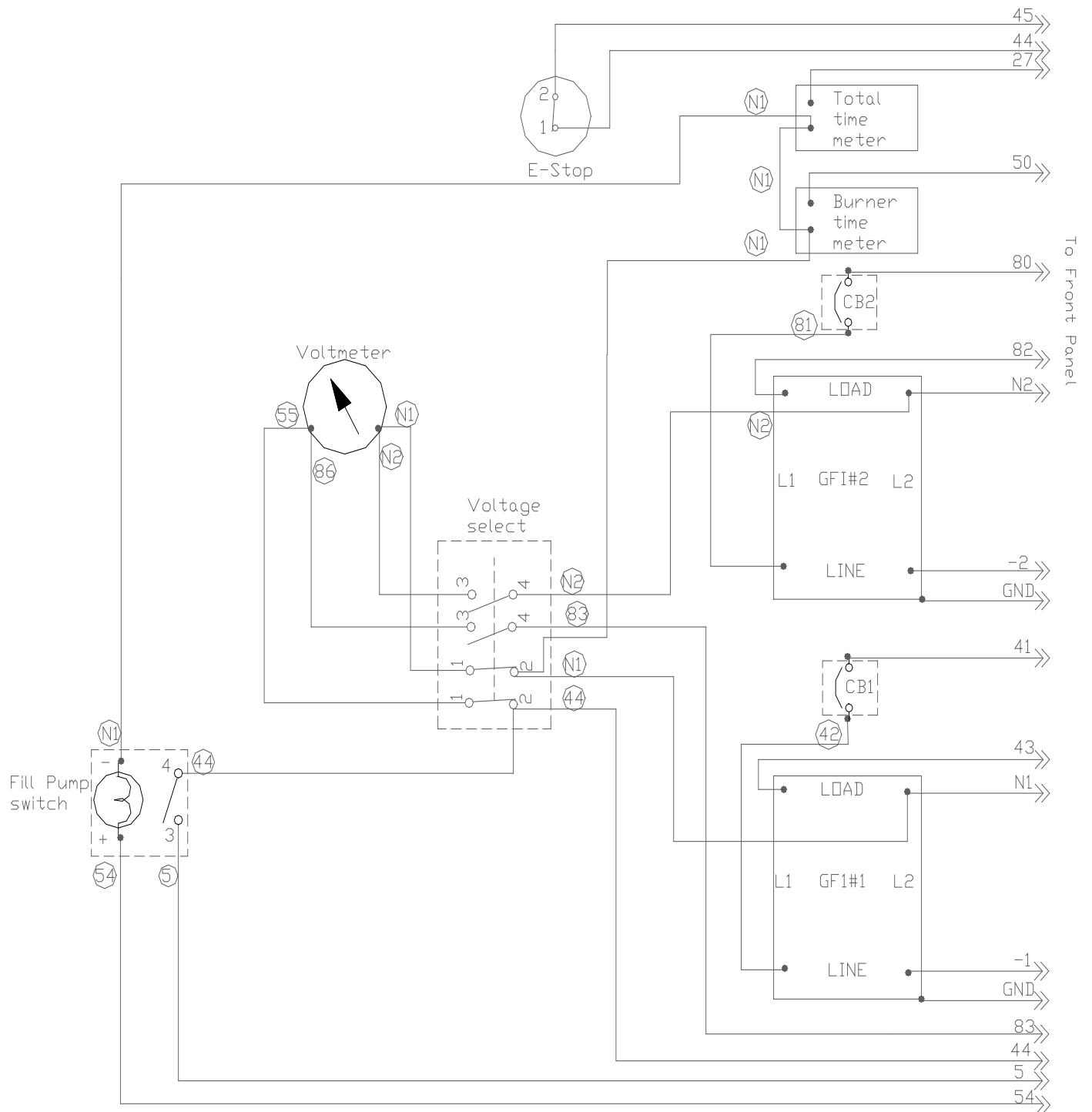
Ref	Part	Description	Qty
1	227065	Bolt, Hex, 5/8 x 4	4
2	174480	Washer, 5/8	8
3	445020	Bearing, Pillow Block	2
4	176260	Washer, Lock, 5/8	4
5	227872	Nut, Hex, 5/8	4
6	440167	Reel Frame, Weldment	1
7	430950	Pulley, 2AV 25 x 5/8	1
8	134990	Pin, Clevis, 3/8 x 1-3/32	2
9	119229	Yoke End, 1/2-24 UNF	1
10	135790	Pin, Cotter, 1/8 x 1	2
11	445027	Nut, Hex, 1/2 UNF	1
12	440179	Reel, Turnbuckle Shaft	1
13	445026	Yoke End, 1/2-24 UNF - Left Hand	1
14	440180	Pulley Guard, Weldment	1
15	445110	Belt, 5L 500	2
16	496160	Pulley, 2AV 10 x 3/4	1
17	496645	Gear Reducer	1
18	441240	Gearbox Bracket	1
19	432189	Key, 1/2 x 2-1/4	1
20	440161	Reel Shaft	1
21	445070	Lock Ring	2
22	445025	Insert, Bearing, Pillow Block	2
23	440163	Reel Weldment	1

Gear Box Assembly ~ 496645

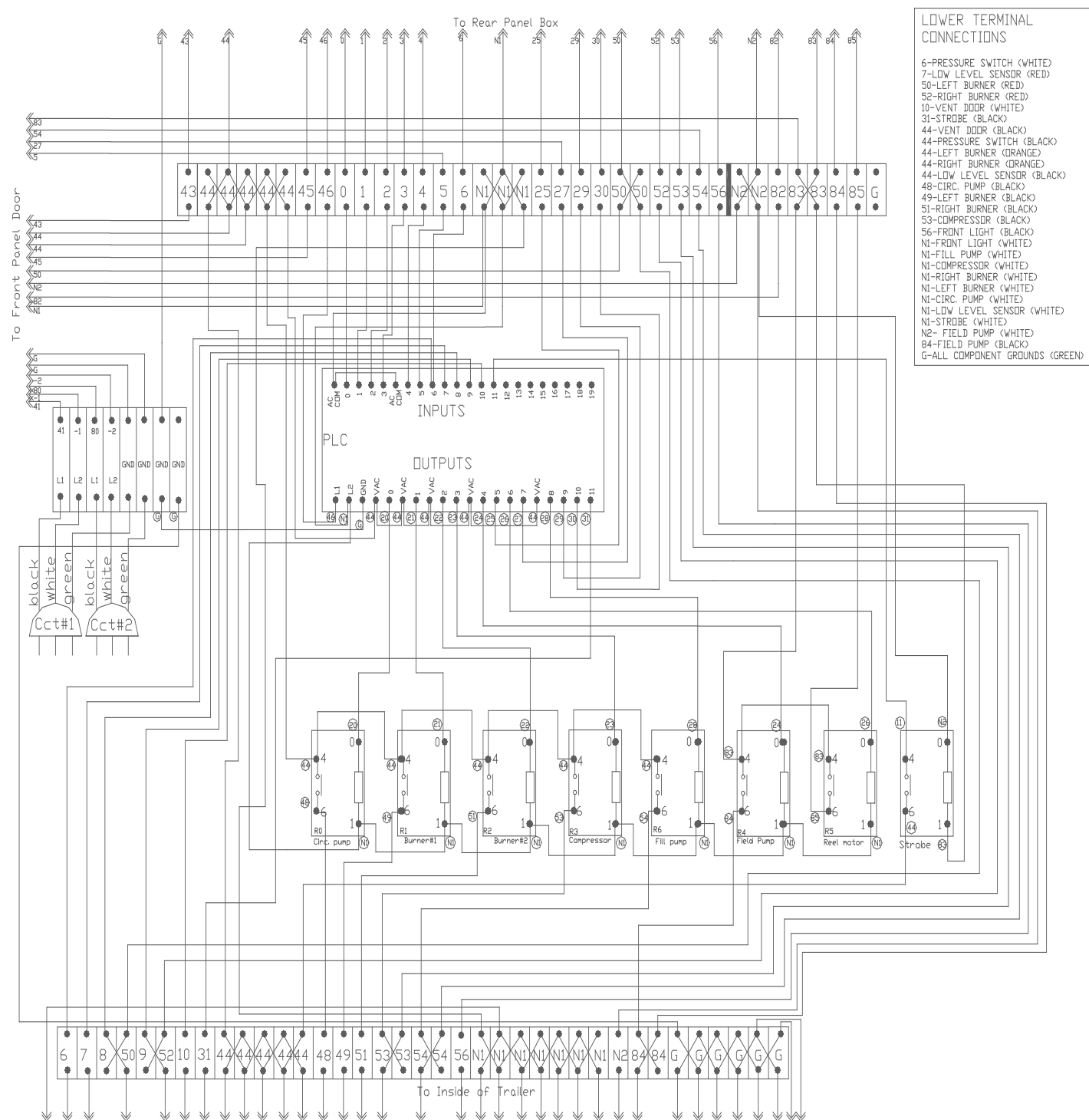


Ref	Part #	Description	Qty
1	199491	Oil Seal, C/R 24897	1
2	227820	Nut, Hex 5/16 UNC	10
3	496320	Case, Front Half	1
4	135470	Spring Pin, 1/4 x 5/8	2
5	110105	Ball Bearing, 6013-2RS	2
6	496642	Sleeve	1
7	432150	Key 3/8 x 1-3/8	1
8	496641	Gear, 60T, 6DP, (10" dia)	1
9	496322	Gear, 60T, 10 DP (6" dia)	1
10	187411	Bushing, Reducer 3/8 x 1/8 NPT	1
11	110550	Breather	1
12	176220	Washer, Spring 5/16	10
13	225980	Bolt, HexHd, 5/16UNC x 1	8
14	496640	Case, Rear Half	1
15	174996	Washer, 3/4 Nom	1
16	227236	Bolt, HexHd, 3/4 x 1	1
17	185630	Plug, Pipe 3/8 NPT	1
18	199380	Oil Seal, C/R 7440	1
19	225990	Bolt, Hex Hd, 5/16 UNC x 1-1/4	2
20	496045	Link, Lever	1
21	185632	Pipe Plug, Magnetic 3/8 NPT	1
22	110410	Ball Bearing, 6204	4
23	432050	Key 3/16 x 1-5/8	1
24	496324	Pinion/Shaft, 12T, 10DP	1
25	496333	Pinion, 12T, 6DP (2" dia)	1
26	432290	Key 1/4 x 2-1/2	1
27	496131	Shaft, Pinion	1
28	128291	Nut, Hex Locking 1/2	3
29	496643	Bolt, Hex Hd, Shear, 1/2UNC x 3.35	3

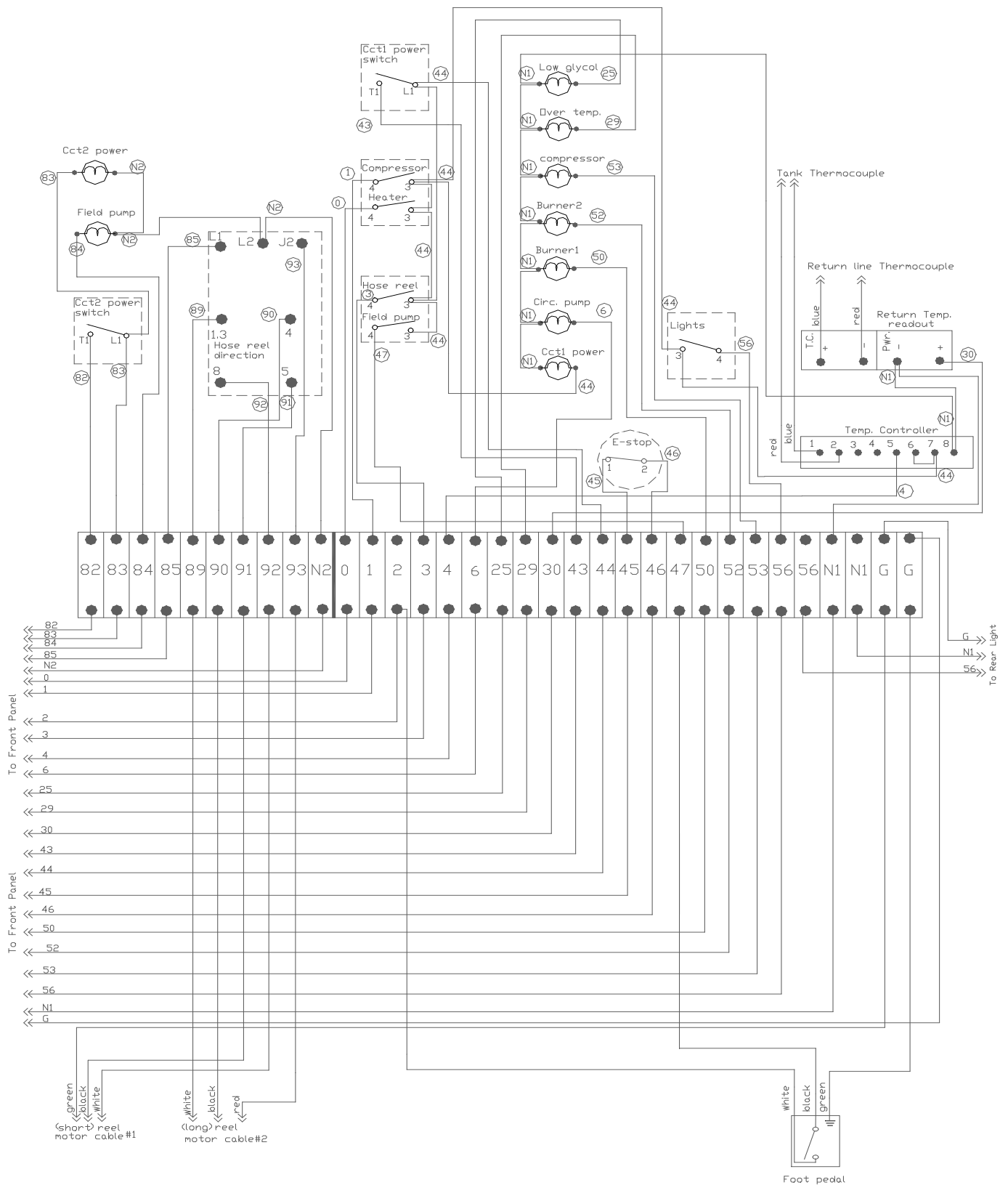
Front Panel Door Schematic



Front Panel Schematic



Rear Panel Schematic



Inside Trailer Schematic

